

Influence of storage on the position of the germinal disc in the fertilized unincubated chicken egg

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ABSTRACT The breeding of male layer chickens is currently considered to be highly uneconomical. In Germany alone, 40 to 50 million newly hatched male chickens were killed annually immediately after hatching. Therefore, it is necessary to find a method for sexing chickens early in the embryonic development, preferably before incubation. The genotypic sex of an egg can be determined using information found in the germinal disc, so knowledge of the exact position of the germinal disc is essential for further sexing, or for other actions such as the *in ovo* injection of agents. Previous studies have shown that the germinal disc is located somewhere on top of the yolk. However, no studies have yet been performed that investigate the influence of time spent in horizontal storage on the position of the germinal disc. Magnetic resonance imaging was chosen to determine

this influence on the position of the germinal disc. It was found that eggs placed horizontally for long periods of time before scanning had significant changes in the positions of their germinal discs compared with those of eggs scanned minutes after positioning. The position of the germinal disc in eggs, minutes after horizontal positioning, deviated 14.7 ± 0.6 mm from the maximum vertical plane of the egg (zero position) in the z-direction; eggs scanned after 96 h of horizontal positioning showed a deviation of only 4.9 ± 1.6 mm. The x-axis also exhibited changes in the position of the germinal disc over time. Immediately after horizontal positioning, the eggs showed a deviation of 0.4 ± 0.4 mm in the x-direction, whereas the deviation after 96 h was 2.9 ± 0.5 mm. These results show that horizontal positioning of the egg hours before the measurement is necessary.

Key words: germinal disc, egg, storage, magnetic resonance imaging

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INTRODUCTION

To avoid the questionable and ethically alarming killing of male layer chickens, it is necessary to identify the gender of the chicken at an early point of time during embryonic development, indicated as stage 1 in Hamburger and Hamilton (1951) or, more precisely, as stage X in Eyal-Giladi and Kochav (1976). With available technology, the genotypic sex of an egg can be determined by examining the germinal disc. The DNA content of this disc is 2% higher in male chickens than in female chickens, caused by a difference between the Z and W sex chromosomes, so a differentiation in the cellular stage is possible (Tiersch, 2003).

The most important fact is that pre-incubation sexing requires accurate access to the germinal disc. Our first analysis with windowed eggs failed to locate the

germinal disc in the region described by Klein et al. (2002), which was approximately 3 mm from the axis of the largest diameter of the egg in the x-direction (defined as the zero position, Figure 1). Because the albumen is quite viscous, it is possible that the germinal disc may need a certain period of time to settle into position. The intention of this study was to determine the dependence on the storage conditions of the mean position of the germinal disc and its variation. The results can change the conditions for storing the eggs before performing chicken sexing to achieve better access to the germinal disc.

Several imaging methods, including ultrasonography, 3D micro CT, optical coherence tomography, and magnetic resonance imaging (**MRI**), have been used to locate the germinal disc in previous studies (Hutchison et al., 1992; Klein et al., 2002; Bain et al., 2007; Bartels et al., 2008). Magnetic resonance imaging is a noninvasive diagnostic imaging modality that generates cross-sectional images of the investigated structures. The most important MRI parameters are echo time (**TE**) and repetition time (**TR**). Echo time is the time between

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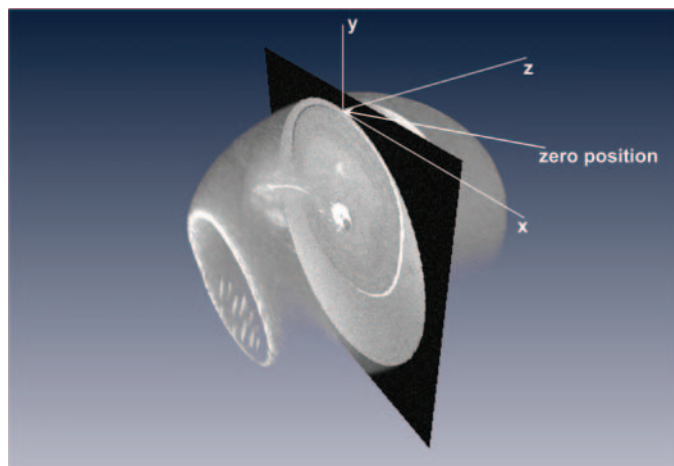


Figure 1. 3D-overview of an egg. The x-, y-, and z-axes are the directions used to determine the deviation of the germinal disc position. The zero position at the largest diameter on top of the egg is used as a reference point. Color version available in the online PDF.

radio-frequency excitation and signal acquisition, and TR is the time between subsequent radio-frequency excitations; TE and TR determine the relative contrast weighting of tissues. A long TR and short TE lead to proton-density-weighted images, as used in the experiments in this study. The image signal in this set up is determined by hydrogen content (Moser et al., 2009). Thus, because of the good contrast and the potential of imaging unopened eggs, we acquired time-dependent magnetic resonance (MR) images to determine the path taken and time scale required for the disc to position itself on top of the yolk, allowing an accurate access for subsequent in ovo chicken sexing. Chicken sexing, not part of this study, is then performed by other methods, such as spectroscopy, that determine the differences in the DNA content (Harz et al., 2008; Steiner et al., 2010).

MATERIALS AND METHODS

Imaging

To image the movement of the germinal disc, a small-animal tomograph, BioSpec 70/30 USR (field strength: 7 T, Bruker BioSpin MRI GmbH, Ettlingen, Germany), was used. A multislice multiecho sequence and a proton-density-weighted contrast with a TE of 17.5 ms and a TR of 1075.4 ms were chosen. The field of view was set to 5×5 cm; the matrix was 256×256 pixels; and the slice thickness was 0.8 mm, with an inter-slice distance of 1 mm. In total, 35 slices for each MR image were taken in a time of 4 min, 35 s.

Experimentation

Magnetic resonance imaging was performed on 5 groups; each consisting of 5 fertilized, unincubated eggs. Fertilized stage-X eggs were obtained from 41-week-old

Lohmann Selected Leghorn chickens. The eggs have a fertilization rate of 95 to 97%; they were stored in normal egg boxes at 21°C (room temperature) for 2 d post-delivery. The first group was positioned horizontally 96 h before the MRI measurements; the second, 72 h before; the third, 48 h before; the fourth, 24 h before; and the last, immediately before measurement. The eggs were fixed with tape in a small animal holder and first scanned in the z-direction (Figure 1). If the germinal disc, the latebra, and the nucleus of Pander were not visible, further scanning in the x-direction followed.

Second, analyses were carried out to examine the influence of opening the egg shell and the eggshell-membranes on the position of the germinal disc and to prove the influence of stress on its position. An image was acquired before the treatment and another image was taken after incising a hole with a diameter of approximately 1 mm on top of the egg at the zero position. The eggshell and the eggshell-membranes were destroyed. Afterward, the hole was enlarged to approximately 3 mm and a third MR image was acquired.

The influence of stress on the position of the germinal disc was also examined by rotating the egg 10 times around the z-axis with a frequency of 1 Hz and further by shaking and moving it for 5 min in several directions. An image was acquired before the treatment and after the treatment; the eggs had to be fixed in the animal holder again, and a second MR image was acquired approximately 5 min after treatment.

Analysis

Image analysis was carried out with the help of Amira 4.1 (Visage Imaging GmbH, Berlin, Germany). The distance of the germinal disc in x-, y-, and z-direction from the zero position was quantified (Figure 1), using an Amira-integrated measurement tool.

Statistical analyses were performed with GraphPad Prism (GraphPad Prism Software Inc., San Diego, CA) and the statistical significance was calculated by a *t*-test.

RESULTS AND DISCUSSION

In Figure 1, the 3-dimensional MR image of an egg shows the x-, y-, z-axis and the zero position used for the analysis. In the cross sections of the MR images (shown in Figure 2), the germinal disc, the latebra, the neck of latebra, and the nucleus of Pander appear brighter; this reflects a higher content of protein or a higher percentage of ionic iron in these parts. The strand which runs upwards from the latebra is called the neck of latebra. The neck of latebra connects the latebra to the nucleus of Pander; the nucleus of Pander contains a mass of cytoplasm, called the germinal disc, from which the embryo begins to develop (Falen et al., 1991). In Figure 2(b), a magnified part of this region is presented, showing the germinal disc, the nucleus of Pander, and the neck of latebra. Because the nucleus

of Pander was not visible in most of the recorded eggs, for further analysis, the neck of latebra was chosen as the orientation to determine the path taken and time scale required for the disc to position itself on top of the yolk, shown in Figures 2(c) and 2(d). A chemical shift artifact shown as a crescent-shaped shadow can also be seen in the MR images, caused by a shift in the signal between fatty and nonfatty tissue and a signal loss at the barrier between fat and water (black arrows Figure 2(a); Weishaupt et al., 2009). Protons with different precession frequency were displayed in another place along the direction of frequency. This chemical shift allows a differentiation of fatty tissue from nonfatty tissue, but does not affect the analysis in this study.

Immediately after horizontal positioning, the germinal discs in the eggs showed a deviation of 14.7 ± 0.6

mm from the zero position in the z-direction, whereas the eggs stored for 96 h had a deviation of 4.9 ± 1.6 mm along the z-axis. In the y-direction, the behavior of the germinal disc is the same: immediately after horizontal positioning, the deviation reached 12.6 ± 1.8 mm, and 96 h later it was 1.1 ± 0.2 mm. In the x-direction, the germinal discs showed a deviation from the zero position of 0.4 ± 0.4 mm immediately after horizontal positioning; in the eggs stored for 96 h, a deviation of 2.9 ± 0.5 mm was shown (Figure 3). A significant change in the position of the germinal discs in x- and y-direction was also detected between 24 h and 96 h. After 24 h, the germinal discs deviated 5.7 ± 0.3 mm along the x-axis from the zero position and 4.6 ± 1.7 mm along the y-axis. Therefore, a storage time of 96 h was necessary to find the germinal discs at the top of the yolk. The deviations of the germinal discs in the z- and y-directions decreased over the period of time studied, whereas in the x-direction, the deviations of the germinal discs showed an oscillation (Figure 3). Figures 2(c) and 2(d) demonstrate the time-dependent behavior of a germinal disc. After 72 h, the germinal disc reached a lower oblique position than it had 72 h before. Klein et al. (2002) show that the mean position of the germinal disc deviates 3.0 ± 2.1 mm in x-direction and has a mean difference to the inner surface of the shell of 2.7 ± 1.1 mm. The eggs were screened between 1 and 7 d after being laid. No detailed information about the time and modality of storage was mentioned. The eggs in the study by Klein et al. (2002) might have been stored in horizontal direction for 48 h. The results of 3.0 ± 2.1 mm in the x-direction and 2.7 ± 1.1 mm in the y-direction correspond to our results after 48 h, 3.6 ± 0.9 mm in the x-direction and 2.2 ± 0.5 mm in the y-direction. In the z-direction, Klein et al. (2002) find the germinal disc in the intact egg ± 2 mm to the $z = 0$ slice, using the slice thickness for the measurement. No other details are mentioned. Our results showed that the deviation of the position of the germinal disc in the z-direction is minimized after 96 h to 4.9 ± 1.6 mm.

Additionally, the influences of stress and opening the eggshell were examined. The thick layer of albumen prevented fast movements of the germinal disc because of its viscous properties. Slow movements have not resulted in a significant change of the position of the germinal disc. Klein et al. (2002) mentioned that the germinal disc is not able to float to an exact vertical position when the egg is positioned with the blunt end upwards. In the vertical position, the chalazae seem to restrict the free movement of the yolk. In the study of Klein et al. (2002), eggs transferred in the horizontal direction over short distances do not alter the position of the germinal disc. The intact layer of thick albumen stabilizes the yolk and, therefore, the position of the germinal disc. Thus, our results correspond to the observations of Klein et al. (2002). Furthermore, opening the eggshell with holes of different diameters showed no visible variation of the position of the germinal disc

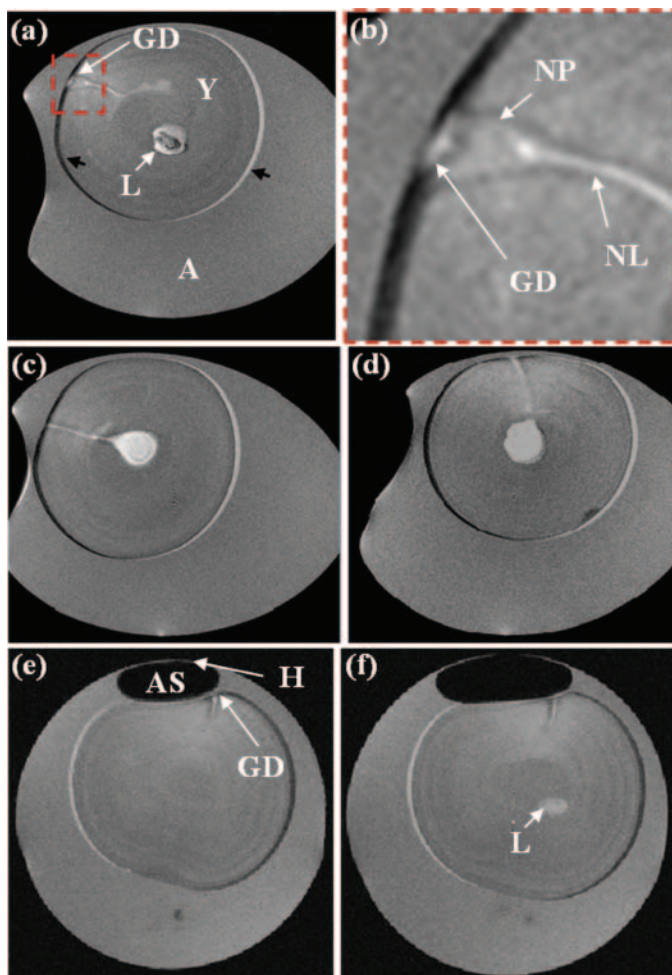


Figure 2. MRI images. (a) Cross section through the z-direction in a fertilized unincubated egg. Black arrows indicate the chemical shift artifact, shown as a crescent-shaped shadow. The dashed box indicates the area shown in (b). (b) Magnified cross section in the region of the germinal disc. (c) and (d) show time-dependent behavior of an egg: (c) egg immediately after horizontal positioning and (d) egg 72 h after horizontal positioning. (e) and (f) show behavior after windowing the eggs with a hole with a diameter of (e) 1 mm and (f) 3 mm. A: albumen, AS: air space, GD: germinal disc, H: hole, L: latebra, NP: nucleus of Pander, NL: neck of the latebra, and Y: yolk. Color version available in the online PDF.

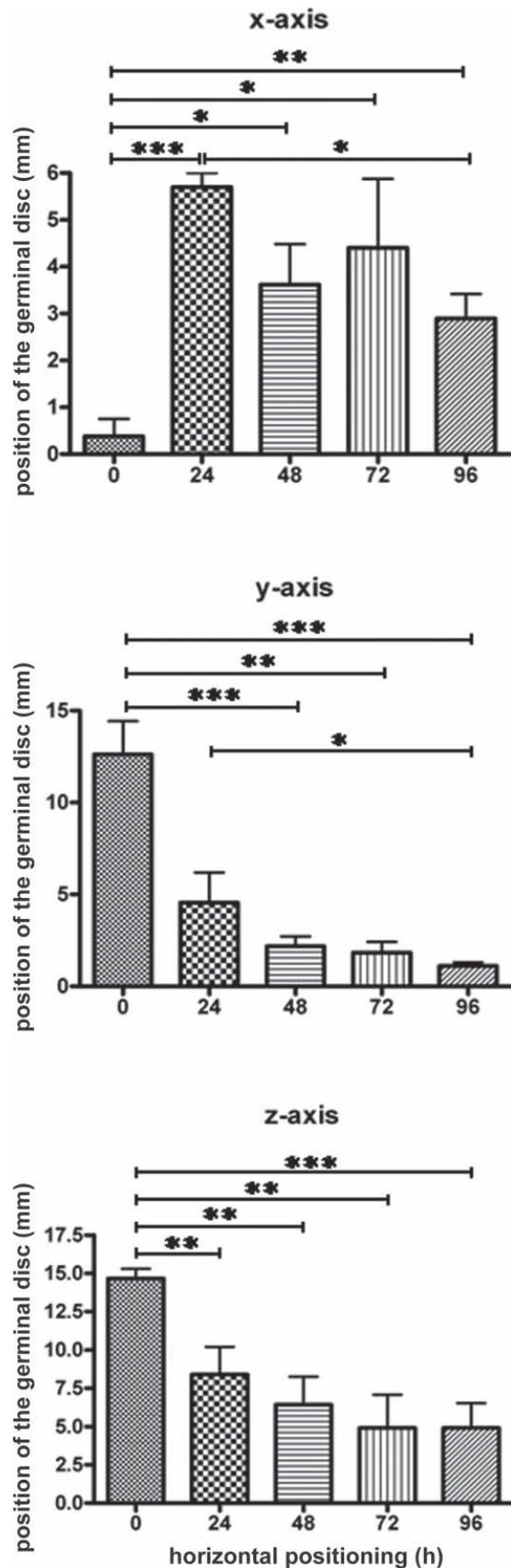


Figure 3. Deviation of the position of the germinal disc in the egg (mean $\pm$ SEM) in millimeters with specification of the significance level, along the different axes from the zero position, after varying amounts of storage time in the horizontal position. Refer to Figure 1 for explanation of the x-, y-, and z-axes. The * indicates $0.01 < P < 0.05$, ** indicates $0.001 < P < 0.01$, and *** indicates $P < 0.001$.

in the x- and z-directions, but an increase in air space was observed and so the distance of the germinal disc in y-direction, to the inner side of the eggshell, was increased [Figures 2(e) and 2(f)].

Conclusions

Magnetic resonance imaging is suitable to locate the germinal disc in the chicken egg and was used in this study to investigate the time required for the germinal disc to locate itself on top of the yolk. This topic has not been described before, and is important for an exact localization of the germinal disc and for further applications such as chicken sexing. For industrial application, MRI is not practical because of the high costs and the long recording times, but it provides convenient information for fundamental research.

The behavior of the germinal disc depends on the manner of storage. In the horizontal position, the germinal disc quickly returned to its previous position on top of the yolk after a short reorientation because the intact layer of thick albumen and the chalazae stabilized the yolk and, therefore, the position of the germinal disc. However, in a vertical position with the blunt end upwards, the chalazae and the thick albumen restricted the free movement of the yolk and, hence, the orientation of the germinal disc on top of the yolk. Thus, this study showed that storage of the eggs in horizontal position for 96 h before the localization of the germinal disc is essential for locating the disc.

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